Comparison of SFDA and FDA Regulations Pertaining to LC/MS Instrument Software Validation

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In today's global business environment, more and more companies are realizing they must focus on multiple marketplaces in order to be successful.. Taking advantage of costs and regulation, much work that was previously done in the USA or Europe is now being done in Asia, particularly in China and India.

For scientific organizations needing to operate in a compliant fashion it is important to understand the similarities and differences between the regulations, standards, and processes used for the US FDA (Food and Drug Administration) and the Chinese SFDA (State Food and Drug Administration).

The goal of this paper is to compare SFDA and FDA regulations pertaining to computer system validation (CSV) of software used to operate Liquid Chromatography-Mass Spectrometry (LC/MS) instruments. In detailing the requirements and regulations of both organization, a comparison of SFDA to FDA regulations can be made.

LC/MS instruments are used extensively for drug discovery, forensic analysis, food quality analysis and other safety-critical research by pharmaceutical companies, research laboratories, universities, hospitals, and testing facilities around the world.

CHINESE STATE FOOD AND DRUG ADMINISTRATION (SFDA)

Since 2003, the SFDA has been under the purview of the State Council of the People's Republic of China, which is responsible for supervision of food and drug safety in mainland China. (State Food and Drug Administration, 2011). The main responsibilities of the SFDA are to develop policies for the discovery, manufacturing and distribution of drugs, Traditional Chinese Medicines (TCM), medical devices, health food, cosmetics, and food at restaurants, cafeterias, and other eating establishments. These policies include:

- Development of national standards for food and drug safety
- Licensure and registration of drug manufacturers
- Supervision of the quality, safety and technical aspects of food and drug research, production, distribution, and use
- Investigation and punishment of illegal activities regarding food and drug safety
- International exchanges and cooperation with other food and drug safety regulations (SFDA, 2011).

Based on these policies, the Drug Administration Law was revised in 2001 and the Ministry of Health Decree No. 79 was issued. Both provide key information related to software validation.

DRUG ADMINISTRATION LAW PEOPLE'S REPUBLIC OF CHINA (PRESIDENTIAL ORDER NO. 45)

The Drug Administration Law was revised in 2001. This law strengthened drug administration to ensure drug quality and safety for human beings. Articles relevant to software validation include Article 7 and 9.

Article 7 of the Drug Administration Law establishes that drug manufacturers must register with the Administrative Department for Industry and Commerce and obtain a Drug Manufacturing Certificate. No one may manufacture drugs without the certificate. To be licensed to sell their drug products in China, all Chinese pharmaceutical companies were required to obtain these certificates from SFDA by June 30, 2004. About 3000 companies met the deadline (SFDA Agent, 2011).

Article 9 requires that drug manufacturers conduct production in accordance with Good Manufacturing Practice (GMP). Ministry of Health Decree No. 79 provided more guidance for this.

MINISTRY OF HEALTH DECREE NO. 79

The Ministry of Health adopted Good Manufacturing Practice (GMP) on October 19, 2010. It was retroactive to March 1, 2010. This decree states that manufacturing and quality management of drugs shall be per GMP. Manufacturers must establish a quality management system covering all factors that influence quality of drugs. GMP shall be strictly implemented with integrity. Any falsification and fraud is forbidden.

Article 241 of this decree is the only article that specifically addresses computer software. And it only discusses the need for change control. However, Chapter 7 addresses qualification and validation in detail.

QUALIFICATION AND VALIDATION

Chapter 7 of the Ministry of Health Decree 79 addresses qualification and validation. It contains Articles 138 through 149.

The manufacturer should identify qualification or validation methods to prove that critical operations can be controlled effectively (Article 138). Software used to run an LC/MS instrument is critical to operations and thus apparently covered by Chapter 7. The scope and extent of these qualification or validation methods should be determined through risk assessment (Article 138). This risk assessment approach is the crux of Good Automated Manufacturing Practices (GAMP®) 5.0 to be discussed in the subsequent section, GAMP® 5 Risk-based Focus.

Article 139 says that the premises, facilities, equipment (such as LC/MS instruments and the software used to run them), and testing instruments should be qualified and this validated state should be maintained.

As it relates to equipment, Article 140 addresses the need for records to be generated and kept as evidence of the following:

1. Design qualification to verify that the design of the equipment is suitable for the intended use
2. Installation qualification to verify that the equipment has been built and installed in accordance with design specifications
3. Operational qualification to verify that the equipment operates in accordance with design specifications.
4. Performance qualification to verify that the equipment can consistently meet performance specifications when operating under normal conditions.

Before any new manufacturing formula or process is adopted, its suitablity in regular production should be validated in advance (Article 141). A cleaning method validation should be performed in order to confirm the effectiveness of a cleaning procedure. This is to prevent contamination and cross-contamination (Article 143).

About change control, Chapter 7 states that qualification or validation should be performed when there is a change in major factors influencing the product quality (Article 142).

In Chapter 7, the requirements of retrospect validation and re-validation also be addressed, which states that: Qualification and validation are ongoing processes. After initial qualification and validation, re-qualification or re-validation may be carried out according to the product quality review. Additionally, critical manufacturing processes and operation procedures should be revalidated at defined intervals to ensure the intended outcome (Article 144).

Validation master plan (VMP) also be noted which states in Article 145 and Article 146. The requirements need to be defined in the validation master plan or other relevant documents to maintain the consistent status of premises, facilities, equipment, testing instruments, process, operation procedures and testing methods.

The qualification or validation protocol should be specific to the equipment or system being validated. The protocol should be reviewed and approved before being executed (Article 147). The qualification or validation be implemented in accordance with a predefined and approved protocol. Moreover, the qualification or validation shall be recorded with sufficient evidence. Upon completion of qualification or validation, a report of the result and conclusions should be prepared, reviewed, approved and archived (Article 148).

Article 149 states that written documents of the process and procedure operations should be established .

It is evident that Chapter 7 provides clear guidelines for qualification and validation of equipment such as LC/MS instruments and the software used to run them. The next section reveals that this SFDA guidance closely resembles the guidelines developed by the USA FDA.

KEY POINTS FOR SFDA INSPECTION OF GLP-LABORATORY

In November 2011, at the 2nd China Computerized Systems Validation Workshop, Dr. Ma presented key points for SFDA inspection of GLP laboratories. These points are summarized in this section.

When inspecting a computerized system, the inspector must carefully consider the data, computer hardware and software, the network and the management processes. It is important to define what constitutes original or "raw" data. There must be an audit trail to record the creation and modification of data. The retention period for data must be decided. Normal and emergency backup and recovery plans must be developed.

Computer hardware and software and network connections must be validated. The system must be configured to meet design specifications so it functions as expected. Then the system configuration must be checked for authenticity, accuracy, reliability, stability and security.

Management processes must be recorded in written procedures often called Standard Operating Procedures (SOP). Personnel must be assigned clear responsibilities and trained to meet these responsibilities. Management personnel must provide appropriate oversight.

Validation checks must ensure that there is a clearly defined process based on good practices such as GAMP. The scope of validation should base on the user requirements and must be well documented in a comprehensive validation plan. The validation plan should include provisions for risk assessment, design qualification, installation qualification, operational qualification and performance qualification. The test acceptance criteria should be pre-defined. There should be a traceability matrix matching requirements to actual tests. The validation plan must also make sure that SOPs are in placed before system put into use.

After completion of validation, a validation summary report should document findings. Change control procedures should be put into place so any change after validation can be recorded. The system should be reviewed periodically including inspection of the original system and any changes.

Common challenges found during SFDA validating a computerized system are listed below:

- Test results are limited to "pass" or "fail" without adequate objective evidence
- Test data not reflective of daily operations. Generation of test data must be carefully undertaken to ensure that the test data are accurate and reflect daily operations
- Too many tests. Tests should be well-crafted to expedite the validation.
- Scheduling conflicts between research or work deadlines and the time needed for validation (Ma, 2011)

UNITED STATES FOOD AND DRUG ADMINISTRATION (FDA)

The FDA considers software validation to be *"confirmation by examination and provision of **objective evidence** that software specifications conform to user needs and intended uses, and that the **particular requirements** implemented through software can be consistently fulfilled."* (General Principles of Software Validation, 2002).

Objective evidence is generally kept in the form of printed screen captures or printed reports. Particular requirements are typically devised via a GAMP 5® risk-based approach. They are recorded in the User and Functional Requirement Specification (UFRS) document. Test protocols are developed to test each requirement. The "confirmation of conformity" to user needs and intended uses is obtained by executing these test protocols and comparing actual system performance against expected results. The point is to demonstrate that the software satisfies its intended use.

It is also important to demonstrate that validation activities were properly planned and that the tests were executed according to the plan.

In the USA, industry groups often develop guidance to meet government regulations. Thus the International Society for Pharmaceutical Engineering (ISPE) developed GAMP® to support compliance with FDA regulations.

GAMP® 5 RISK-BASED FOCUS

GAMP® stands for Good Automated Manufacturing Practice (ISPE, 2010). ISPE has also issued guidelines specific to the pharmaceutical industry called The Good Automated Manufacturing Practice (GAMP®) Guide for Validation of Automated Systems in Pharmaceutical Manufacture. These two guides describe a set of principles and procedures that help ensure that pharmaceutical products have the required quality. So far there have been five revisions of GAMP®. GAMP® 5 is the most recent.

The most important update in GAMP® 5 is the focus on risk management as it relates to human safety (General Principles of Software Validation, 2002). GAMP® 5 requires validation if there could be an impact on human safety, product quality, or data integrity (GAMP 4® to GAMP 5® Summary, 2008).

When doing risk assessment in accordance with GAMP® 5, it is important to assess the probability and severity of harm if a fault occurs, rather than the probability of the fault occurring. Therefore, risk must be scrutinized using the criteria of whether the functionality is critical or not. For example, software controlling the acquisition of data for evaluating food quality is more critical than software used to create a report template for pagination. Therefore, according to GAMP® 5 more testing should be done on the data acquisition than report templates.

GAMP 5® describes five steps to performing a quality risk assessment:

1. Perform initial risk assessment and determine system impact
2. Identify functions with impact on product quality and data integrity
3. Perform functional risk assessments and identify controls
4. I mplement and verify appropriate controls
5. Review risks and monitor controls

SOFTWARE VALIDATION DOCUMENTS

The following list of documents reflects a typical software validation document set that would meet FDA requirements. These documents also meet SFDA requirements.

Validation Plan (VP) – comprehensive plan for conducting validation

Validation Risk Assessment (VRA) – risk assessment based on GAMP® 5.0 guidelines

User Functional Requirements Specification (UFRS) – list of user and functional requirements to meet design specifications

System Design & Configuration Specification (SDCS) – description or "blueprint" of the system

Test Plan (TP) – list of tests to be conducted during validation

Design Qualification (DQ) Protocol – verification that software has been designed to meet intended use

Installation Qualification (IQ) Protocol – test scripts to verify that software has been configured to meet design specifications

Operational Qualification (OQ) Protocol – test scripts to verify that software operates in accordance with design specifications

Performance Qualification (PQ) Protocol – test scripts to verify that software consistently meets performance specifications when operating under normal conditions

Traceability Matrix (TM) – method of tracking the user and functional requirements to the test scripts

Validation Summary Report (VSR) – comprehensive summary of validation activities and the results of same

21 CFR PART 11 ASSESSMENT

A 21 CFR Part 11 Assessment is important. 21 CFR Part 11, Electronic Records; Electronic Signatures (Part 11) regulates the security, reliability and integrity of laboratory data, and the security and integrity of electronic signatures. It contains relatively few signature requirements; however, where signatures are required 21 CFR Part 11 defines how an electronic signature must be derived and the meaning of the electronic signature. Many 21 CFR Part 11 requirements will be met with a combination of technical and procedural controls.

COMPARISON SFDA TO FDA

On December 11, 2007, the USA Department of Health and Human Services (HHS) and the Chinese SFDA made an agreement based on these general principles:

> The USA and China will engage in regulatory cooperation regarding the export of drugs and medical devices
>
> The USA and China will share information to improve their mutual understanding of and to gain greater confidence in each country's regulatory system. Each country will share information with the other about relevant laws, regulations, areas of jurisdiction, and public health and safety.
>
> The USA and China will engage in regulatory cooperation regarding improving the authenticity, quality, safety, and effectiveness of drugs, and medical devices (Memoranda of Understanding, 2011).

In this agreement China and the USA decided to hold annual meetings to discuss a phased approach for formal evaluation of drugs and medical devices to be exported based on these factors:

- Risk to public health associated with drug
- Refusal of either country to admit known problems associated with a drug (safety alerts, recalls, enforcement actions…)
- Fraudulent or deceptive labeling
- Advertising of drugs
- Gaps between SFDA and FDA regulations (Memoranda of Understanding, 2011)

It is important to note that in this agreement the SFDA required that manufacturers exporting drugs to the USA must be certified by the SFDA. There was also a provision for expedited inspection of laboratories with or without advance notice (Memoranda of Understanding, 2011). This agreement is an important step for both countries to work together for food and drug safety.

The US FDA recommends a major regulatory inspection/audit every 2 years, whereas the SFDA recommends every 5 years. Additionally, the SFDA inspections and audits are conducted with a greater degree of scheduling and fore-knowledge than with the US FDA.

Research presented in this paper reveals that the current SFDA regulations have much in common with the USA FDA regulations (SFDA Agent, 2011). Table 1 summarizes the comparison. There is also one important difference. No specific guidance was found in the SFDA documents regarding electronic signatures.

While there are many similarities, it is important for software validation efforts to focus on the laws and regulations applicable to the manufacturer who is validating the software used to run the LC/MS instrument.

TABLE 1: COMPARISON SFDA AND FDA FOR REGULATORY COMPLIANCE OF BA LABS

Guideance	SFDA	FDA
Must adhere to Good Laboratory Practice (GLP)	X	X
Focus on risk-based qualification and validation (GAMP® 5)	X	X
Require certification	X	X
Equipment (such as LC/MS instrument) and testing instruments must be qualified. (Note: Software is required to run equipment so it is reasonable to assume that software validation falls into this category.)	X	X
Recommend time between major inspections or audits	5 Years	2 Years
Qualification and validation must be specific to the equipment or system being validated	X	X
Validation required before any new manufacturing formula or process is adopted	X	X
Qualification or validation should occur when there is a significant change	X	X
Qualification or validation are ongoing processes to maintain a validated state	X	X
A validation master plan is required	X	X
Protocols include Design Qualification protocol (DQ), Installation Qualification protocol (IQ), Operational Qualification protocol (OQ), and Performance Qualification pProtocol (PQ)	X	X
Qualification and validation must be in accordance with a pre-approved protocol	X	X
Records must be generated as objective evidence of qualification and validation	X	X
Standard Operating Procedures (SOP) must be established and documented	X	X

BIOGRAPHY

Stacy D. Nelson, CCNA®, PMP® is a software validation specialist who brings her prior software validation experience at NASA to the GLP laboratories around the world. She has published over 25 technical papers about software validation for safety-critical software, as well as, co-authoring a book titled, Independent Verification and Validation of Neural Networks. She recently won the AB Sciex DNA award for Demonstrated Noteworthy Achievement for leadership of Software Validation projects in USA and China. In November 2011, she also passed the HSK Level 1 Chinese Language Proficiency exam.

Jing MA, Ph.D is the Facility Manager at the National Shanghai Center for New Drug Safety Evaluation and Research. Recently, she was an invited speaker at the 2nd China Computerised Systems Validation Workshop 2011 in Shanghai, China. She spoke her experience with jumping hurdles to achieve compliance with the SFDA.

Dave A. Abramowitz, PMP® As the Senior Manager for the AB SCIEX Compliance Services and Validation Consulting organization, Dave has led the group since its inception in 2006. He currently travels the globe to work with AB SCIEX's mass spectrometry users in mapping out and designing compliance strategies. For nine years prior, Dave was with Cisco Systems in various IT and Operations roles, where he achieved his PMP® certification. Dave also has extensive experience in the financial derivatives markets working for various banks and trading firms in New York and Chicago. Dave has a BA in Economics and an MBA from the University of Texas at Austin.

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